Hack the Climate 2026

Curtailment Action Lab

Compare grid options before a planning decision

A planning prototype with explicit numerical checks

The planning decision

Which permitted grid arrangement can accept more wind?

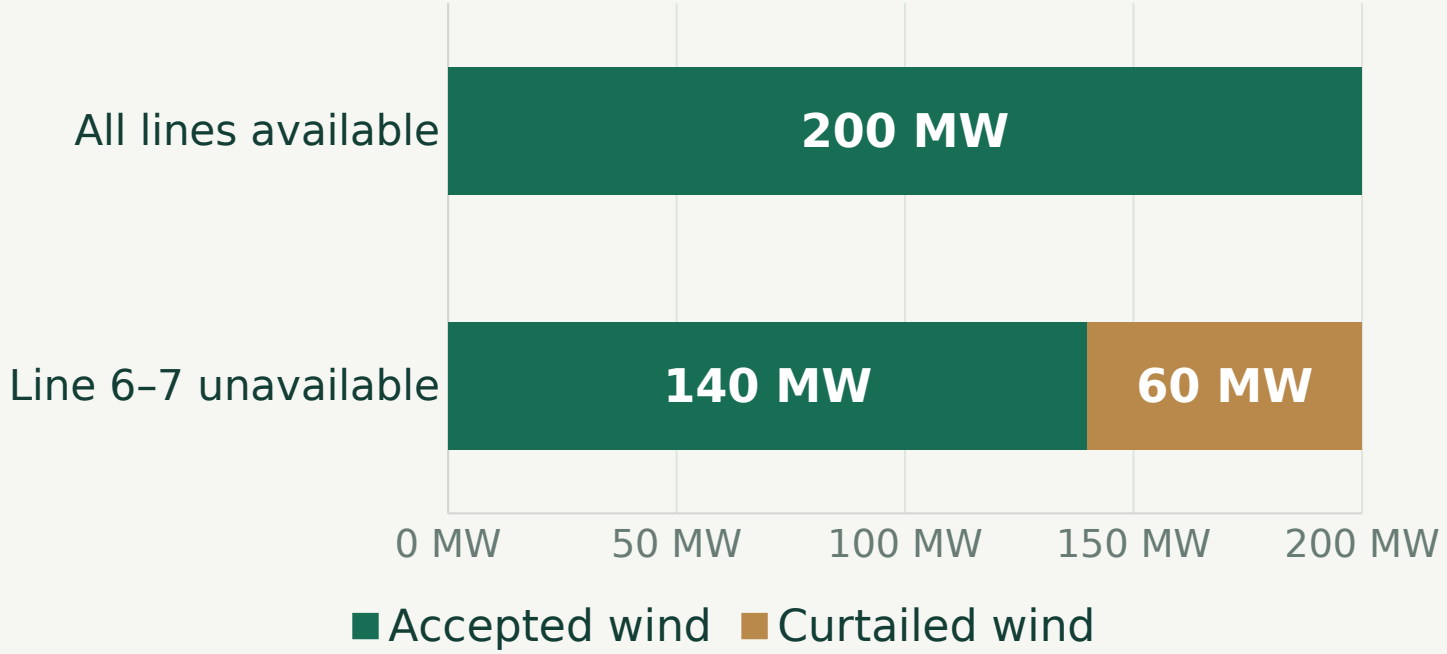
The comparison holds wind availability, consumer demand and equipment limits constant.

User: transmission planning or outage-planning engineer

The planner supplies feasible alternatives and retains approval.

Reference-network result

200 MW available wind, 315 MW consumer demand



60.0 MW

more wind
accepted
in this snapshot

Hypothetical wind at bus 6. Public nine-bus reference network.
Published equipment limits remain unchanged.

The working planner interface

Overview	Purpose and current verification status
Scenario Lab	Run options and compare results
AI Analyst	Inspect requested calculations and their outputs
Evidence	Review checks and export the record

AI proposes the next calculation

1	**Request**	The engineer asks for a comparison.
2	**Tool call**	AI selects an allowed scenario and invokes the solver.
3	**Evidence**	The verifier checks each computed result.
4	**Decision**	The engineer reviews the comparison and its limits.

Azure integration ready. Live AI benefit remains unmeasured.

Evidence and its scope

Profile	Evidence available	Claim allowed
Reference network	21 confirmation cases pass 13 checks per case	Verified constructed scenario Listed steady-state checks
Irish planning model	Public planning data Model controls incomplete	Diagnostic only Irish impact unverified
Microsoft AI	Bounded solver tool calls Local interface tests	Integration ready Live benefit unmeasured

Snapshot results do not establish annual savings, N−1 security or field performance.

A bounded operator pilot

One network area. Operator-approved options.

Input	A verified network case, wind and load assumptions, permitted outages and required security checks.
Measure	Accepted wind against the existing plan, failed options, engineering review time and AI contribution.
Output	An auditable comparison for the planner to review.

Appendix: numerical verification

Power balance	Independent complex branch equations at every bus
Limits	Voltage, both-end thermal loading, generator P/Q and connectivity
Second solution	Independent SciPy power flow compared with PYPOWER
Confirmation	21/21 predefined cases pass. Low-wind control gives zero benefit.
Outside scope	N−1, dynamics, switching sequence, field validation